Four nonanomalous positions: 3-30 Hz

Independent unwrap per position; only a constant whole-curve 360-deg shift at the first plotted frequency. No smoothing.
The anomalous z = -1 mm record is excluded from this focused comparison.

z = -0.5 mm z = 0 mm z = 0.5 mm z = 1 mm

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$

$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$

$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$

$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

Unwrapped phase (deg)

Frequency (Hz)